

System of Linear Equations

A collection of one or more linear equations with same number of unknowns $x_1, \dots, x_n$

e.g. $2x_1 - x_2 + 1.5x_3 = 8$ ①
 $x_1 - 4x_3 = -7$ ②

solutions: $(x_1 = -7, x_2 = -22, x_3 = 0)$, $(x_1 = 1, x_2 = -3, x_3 = 2)$,
solution set:

$$\begin{aligned} \textcircled{1} - 2 \times \textcircled{2} &\Rightarrow -x_2 + 1.5x_3 - (-8x_3) = 8 - (-14) \\ &\Rightarrow -x_2 + 9.5x_3 = 22 \\ &\Rightarrow x_2 = 9.5x_3 - 22 \end{aligned}$$

$$\textcircled{2} \Rightarrow x_1 = 4x_3 - 7$$

Then the solution set is the set of (x_1, x_2, x_3) such that
 $x_1 = 4x_3 - 7$ and $x_2 = 9.5x_3 - 22 \Rightarrow$ infinitely many solutions

A system of equations can have

{	1. no solution	inconsistent
	2. exactly one solution	consistent
	3. infinitely many solution	

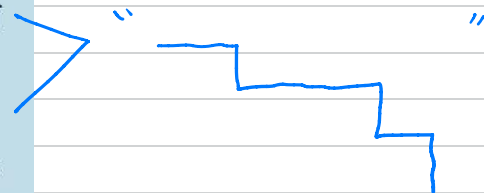
Row Reduction & Echelon Forms

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.



eg.

$$\begin{bmatrix} \textcircled{2} & -3 & 2 & 1 \\ 0 & \textcircled{1} & -4 & 8 \\ 0 & 0 & 0 & \textcircled{5/2} \end{bmatrix}$$

EF

and

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & 29 \\ 0 & \textcircled{1} & 0 & 16 \\ 0 & 0 & \textcircled{1} & 3 \end{bmatrix}$$

EF & RREF

Vector Equations

Linear Combination

Given vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p \in \mathbb{R}^n$ and given scalars $c_1, c_2, \dots, c_p \in \mathbb{R}$, the vector $\vec{y}$

$$\vec{y} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_p \vec{v}_p$$

is called a linear combination of $v_1, \dots, v_p$ with weights $c_1, \dots, c_p$

Span of vectors

If $v_1, \dots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, \dots, v_p$ is denoted by $\text{Span}\{v_1, \dots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated)** by $v_1, \dots, v_p$. That is, $\text{Span}\{v_1, \dots, v_p\}$ is the collection of all vectors that can be written in the form

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p$$

with $c_1, \dots, c_p$ scalars.

The Matrix Equation $A\vec{x} = \vec{b}$

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the product of A and $\mathbf{x}$, denoted by $A\mathbf{x}$, is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights; that is,

$$A\mathbf{x} = \underbrace{\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}}_{\text{vectors}} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \underbrace{x_1}_{\text{weights}} \mathbf{a}_1 + \underbrace{x_2}_{\text{weights}} \mathbf{a}_2 + \cdots + \underbrace{x_n}_{\text{weights}} \mathbf{a}_n$$

$A\vec{x} = \vec{b}$ is consistent $\Leftrightarrow \vec{b}$ is a linear combination of columns of $A \Leftrightarrow \vec{b} \in \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$.

Solution Set of Linear System

Homogeneous system

$$\left[\begin{array}{cccc|c} * & * & - & * & 0 \\ * & * & . & * & 0 \\ * & * & - & * & 0 \end{array} \right]$$

trivial solution $\longleftrightarrow$ pivot in every column $\longleftrightarrow$ no free variable

non-trivial solution $\longleftrightarrow$ some column is not a pivot $\longleftrightarrow$ free variables

Use free variable $\Rightarrow$ parametric vector form.

Linear Independence

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0} \quad (2)$$

e.g. $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

e.g. $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

The columns of a matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution. (3)

$$1 \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \vec{0} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Linear Transformation

$$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

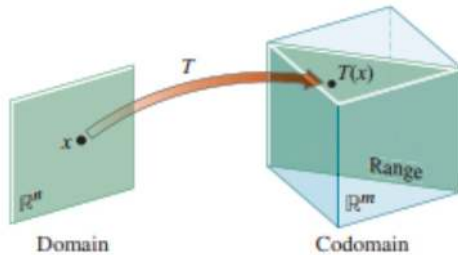


FIGURE 2 Domain, codomain, and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

$$\text{e.g. } T(\vec{x}) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \vec{x}$$

domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^2$

range: $\left\{ \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2 \mid a=0 \right\}$

Matrix transformation

$$T(\vec{x}) = A\vec{x} \quad \text{i.e. } T: \vec{x}_1 \longrightarrow A\vec{x}$$

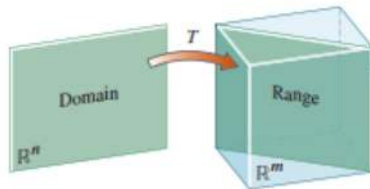
Linear transformation

A transformation (or mapping) T is **linear** if

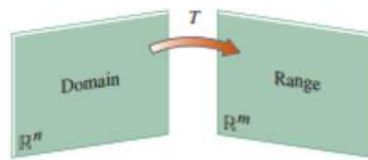
- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
- (ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .

One-to-One & Onto

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.

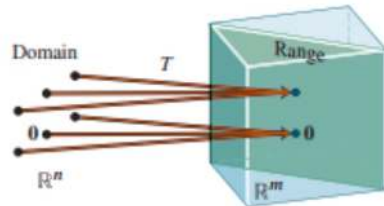


T is not onto $\mathbb{R}^m$

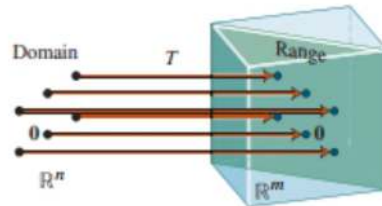


T is onto $\mathbb{R}^m$

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.



T is not one-to-one



T is one-to-one

$A\vec{x} = \vec{b}$ has a solution for any $\vec{b}$

$\Rightarrow$ If row reduce A , we don't want

$$\begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ pivot in every row

$A\vec{x} = \vec{b}$ has unique solution in particular

$A\vec{x} = \vec{0}$ has only trivial solution

$\Rightarrow$ pivot in every column.

THEOREM 10

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

$$A = [T(\mathbf{e}_1) \quad \cdots \quad T(\mathbf{e}_n)] \quad (3)$$

$$\text{If } A = [\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3 \quad \cdots \quad \vec{a}_n]$$

$$T(\vec{e}_1) = \vec{a}_1, \quad \cdots, \quad T(\vec{e}_n) = \vec{a}_n$$

Matrix Operation

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \quad C^2 = C \cdot C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

C & D commute if $CD = DC$
↑ ↑
square matrices